

	User guide on Variant formula update DML
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Revision History:

Rev No.	Author	Date	Description of Change	Revision	Approver
1	Amol S Pujari	24-06-2020	New release	1	Subrata Sen
2	Amol S Pujari, Arjun Paithane	05-05-2022	Functionality extended to inter platform module change	2	Subrata Sen
3	Arjun Paithane	16-09-2025	DMU reviewer node added	3	Subrata Sen

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Introduction

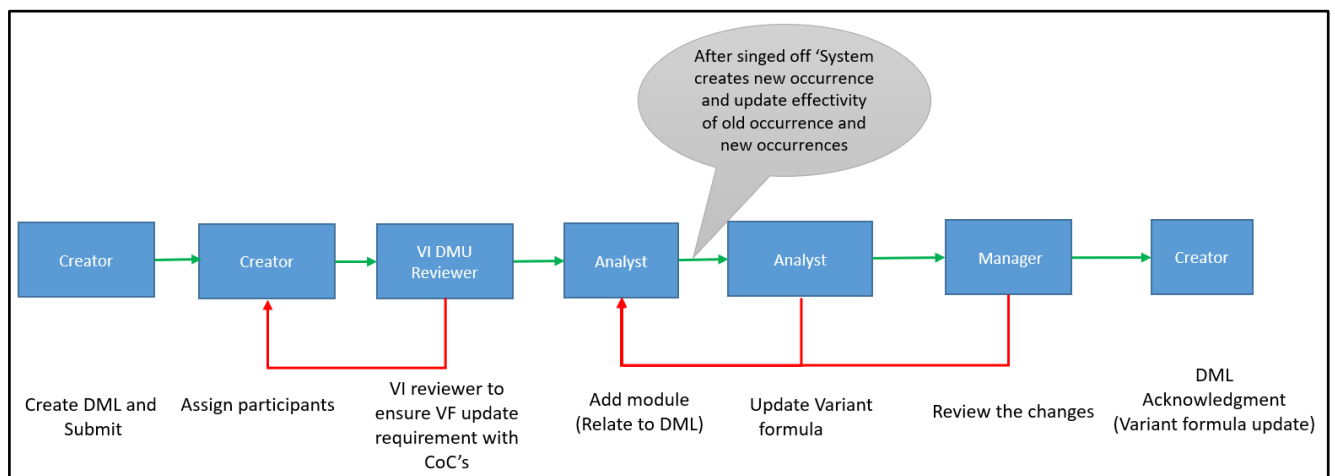
In Modular BOM Platforms, modules are having variant formula, which defines applicability of module. CoC puts the variant formula. Modules undergoes change management due to engineering change or change in variant formula. This document describes workflow for Variant Formula update of Module and occurrence effectivity stamping with respect to variant formula updation.

Rules for variant formula update workflow

- I. CoC will create DML.
- II. System allows to attach only released modules to VF change DML.
- III. Variant condition expression editor will be enabled only for release modules except NR WIP module

DML workflow

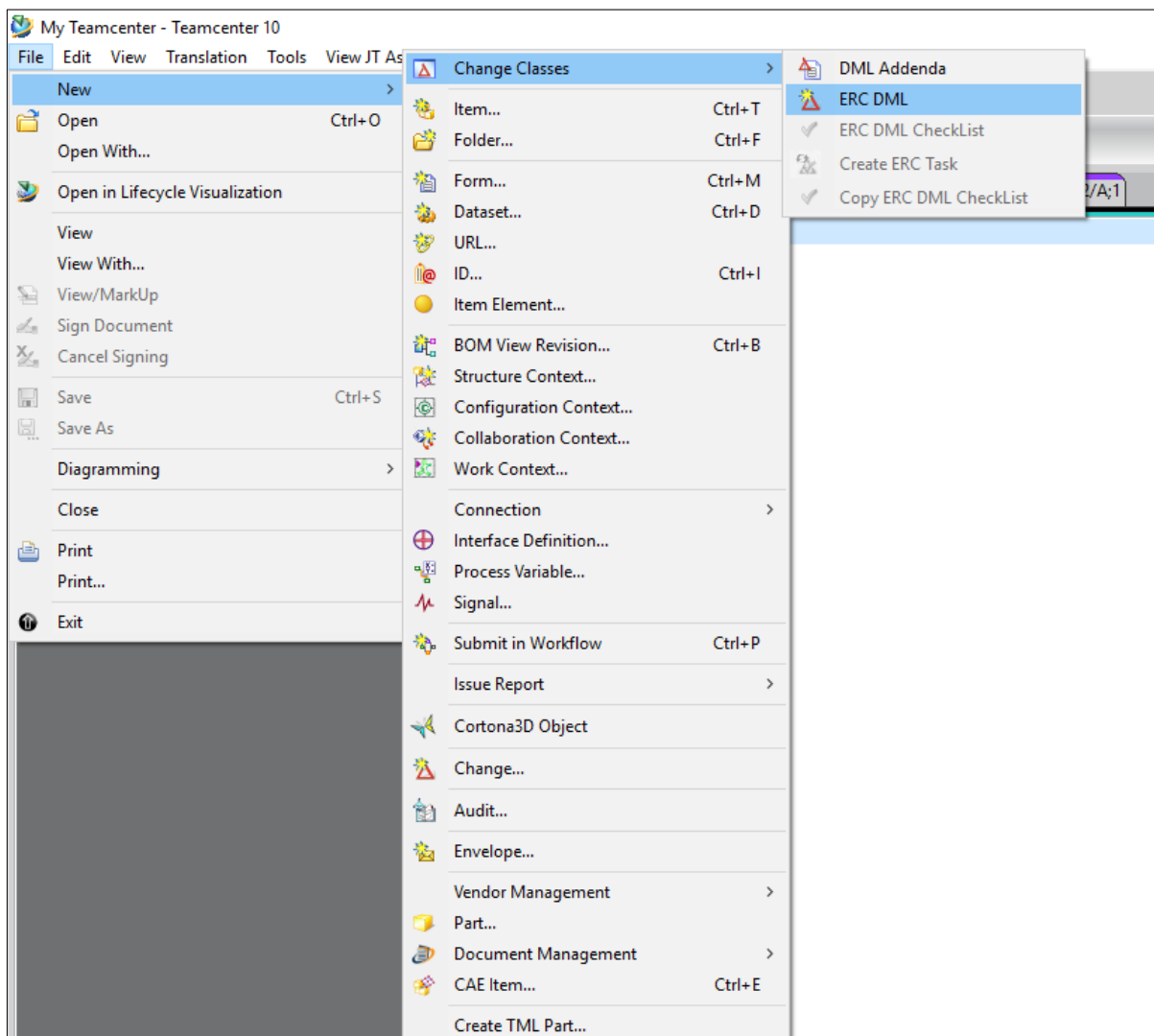
Below is the workflow for Variant formula updation



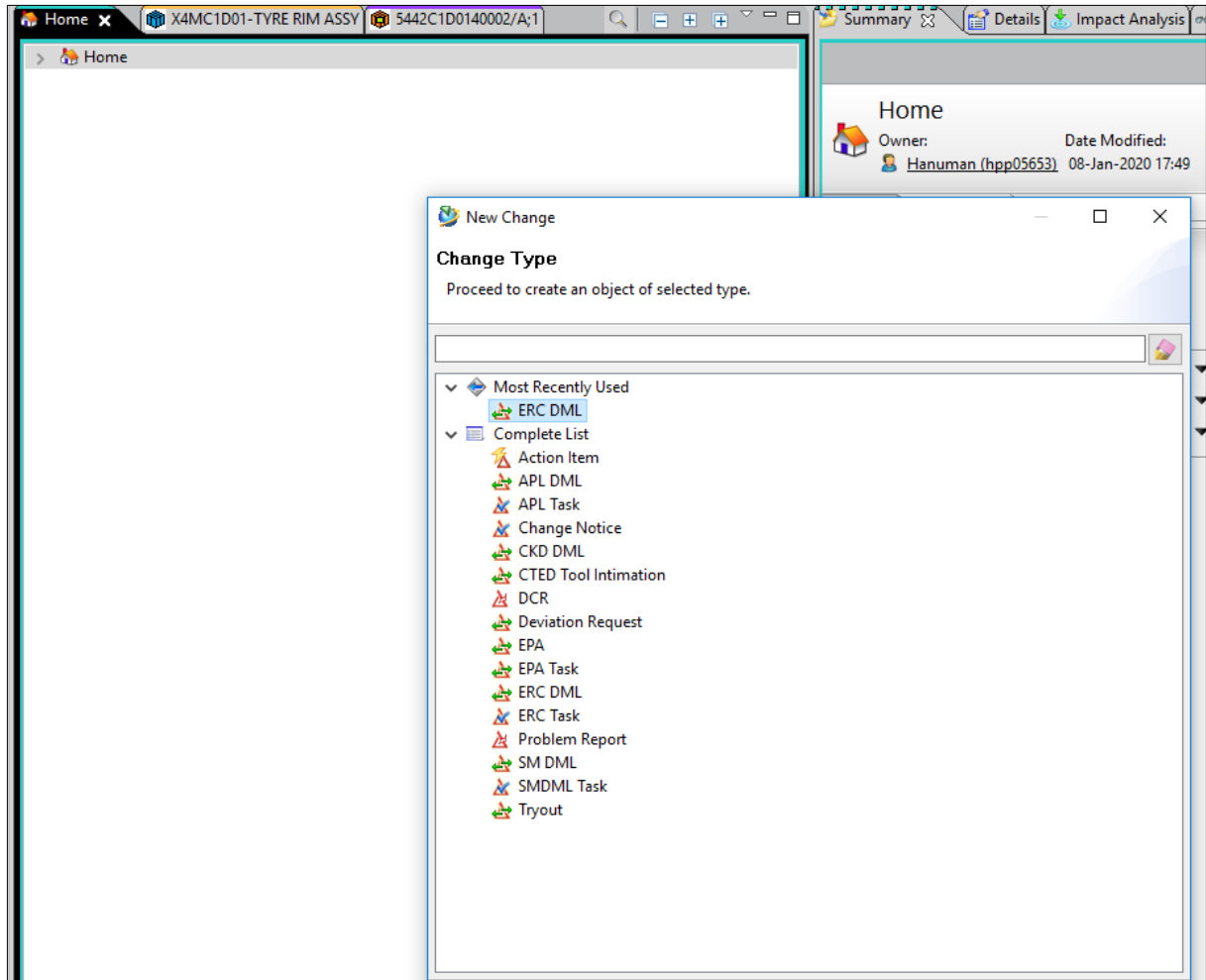
After attaching released module to task system will create occurrence and attach it to respective architecture node. System will update occurrence effectivity on old and new occurrences. Old occurrence shall be stamped with end date New occurrence shall be stamped with start date as current date and end date as future date

Step 1: Create DML and Submit

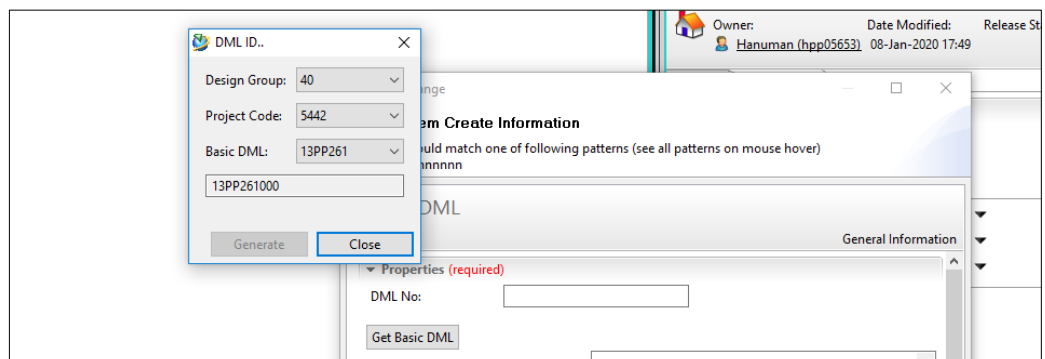
1.1: File > New > Change Classes > ERC DML



1.2: Select 'ERC DML' > Click 'Next'



1.3: DML creation form will open. Create Basic DML (DML id generation) and fill DML details



New Change

Change Item Create Information
Define change item create information

ERC DML

General Information

Reason:* NEW REQUIREMENTS (NEW VC, NEW AGGREG. ▾)

Priority: Mid ▾

Reason Desc: Variant forumula updation

Release Type:* Update Variant Formula ▾

Project Code:* 5442 ▾

Design Group:* 40

Module Address:

Aggregate:

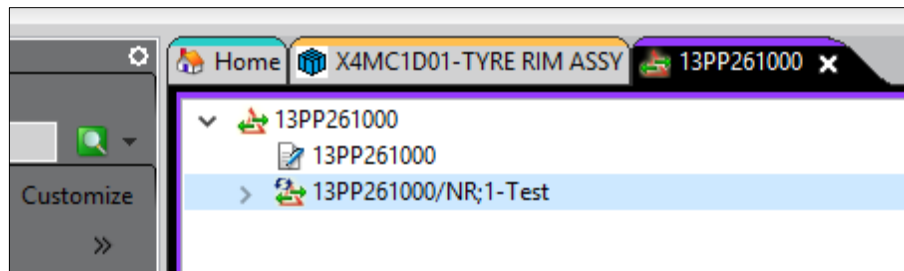
Proposed Solution:

☒ Open On Create

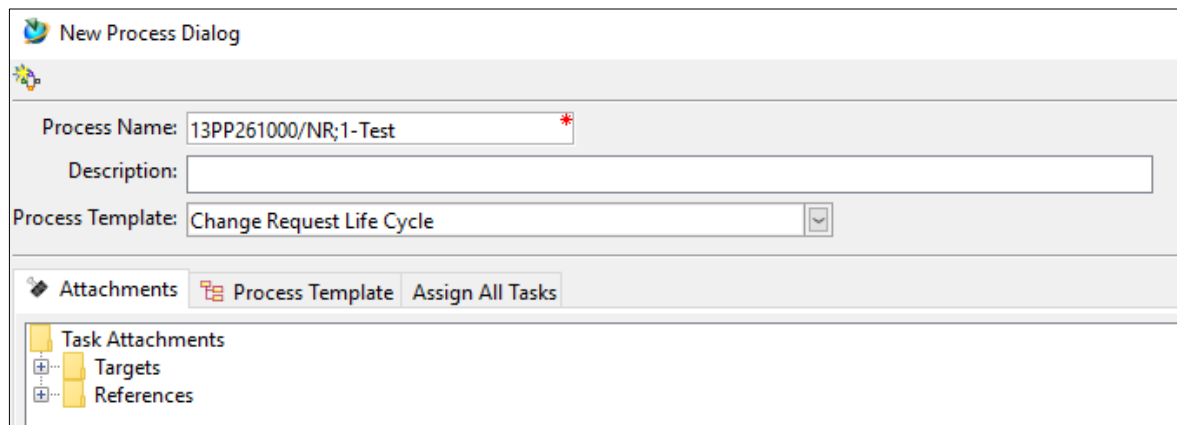
< Back Next > Finish Cancel

Release type- Update Variant Formula
(*) denotes mandatory fields

DML get creates.



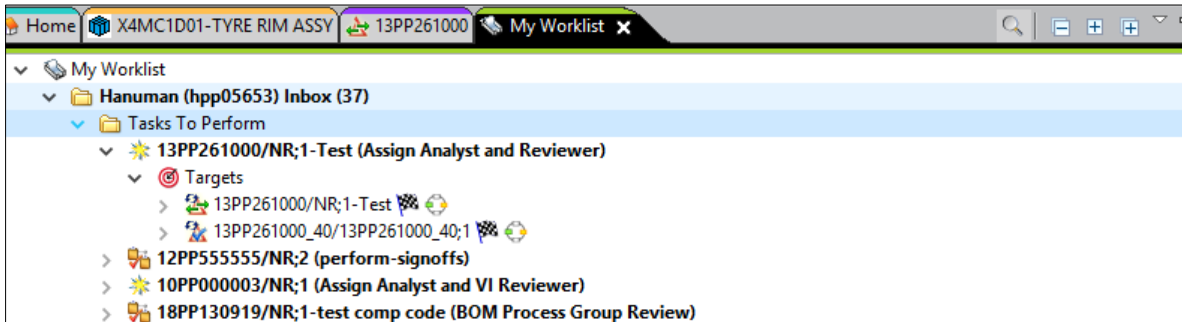
1.4: Submit into lifecycle “Change Request Life Cycle”
Select DML and press Ctrl + P



Now DML submitted into life cycle and available into creator’s worklist
After submitting into lifecycle, life cycle symbol appears in front of DML i.e. 

Step 2: Assign Participants (Assign Analyst and Change Review Board (COC) and VI/DMU Reviewer)

Now DML available with creator's login for assignment 'Assign Analyst and Change Review Board(COC) and VI/DMU Reviewer'.

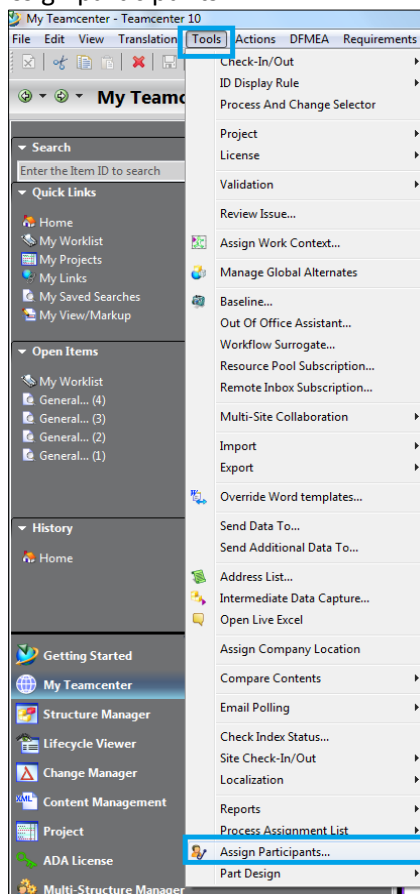


2.1: Assign Analyst

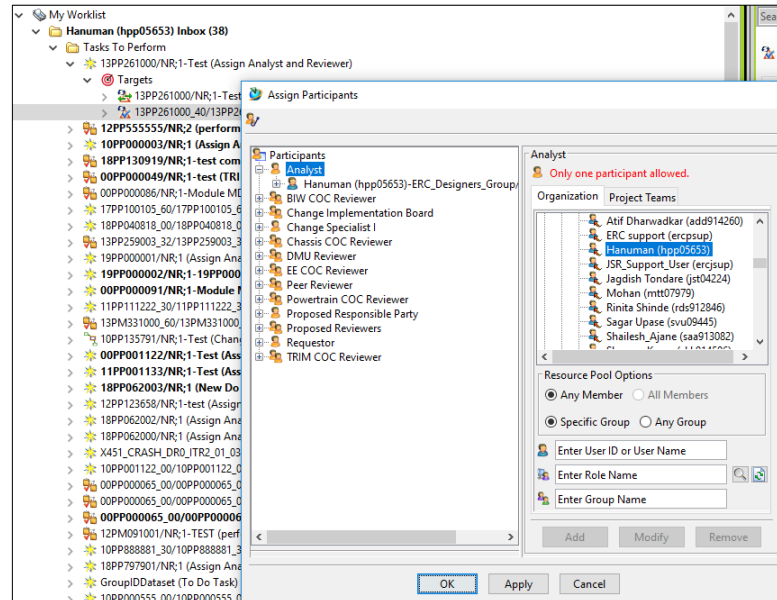
Go to My Worklist of DML Creator - Check Assignment

Select Task from Targets of my Worklist

Tools >Assign participants



Select Analyst from Participants, enter user, search and click on Add

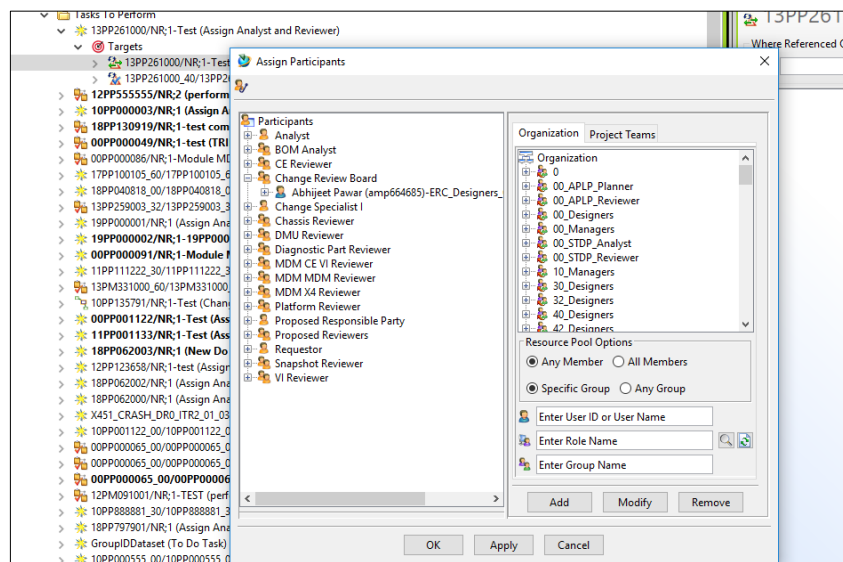


2.2: Assign Change review Board (Manager)

Go to My Worklist of Designer - Check Assignment

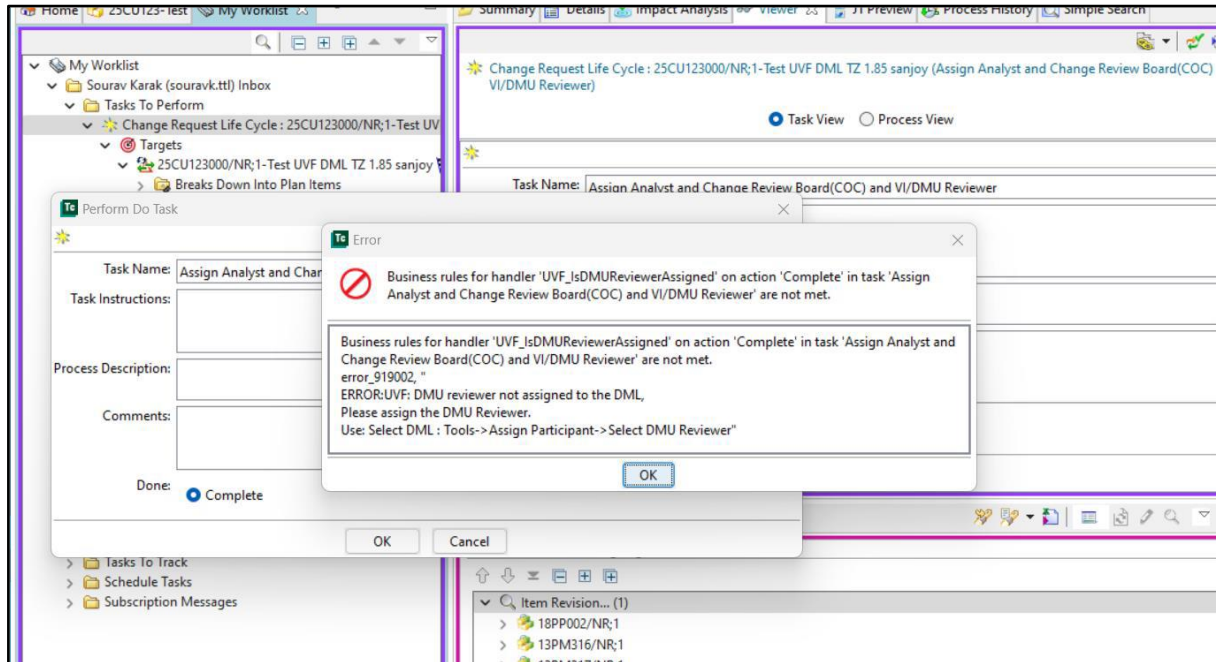
Select DML (not task) from Targets of my Worklist

Tools >Assign participants

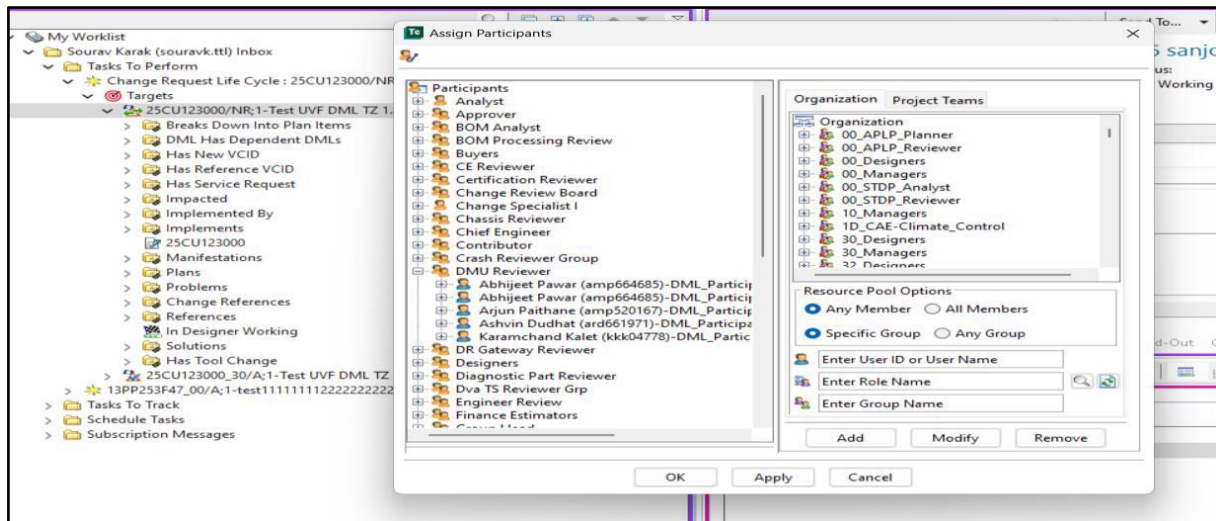


2.3: Assign VI DMU Reviewer

Below validation will come if DMU Reviewer is not assigned for UVF DML

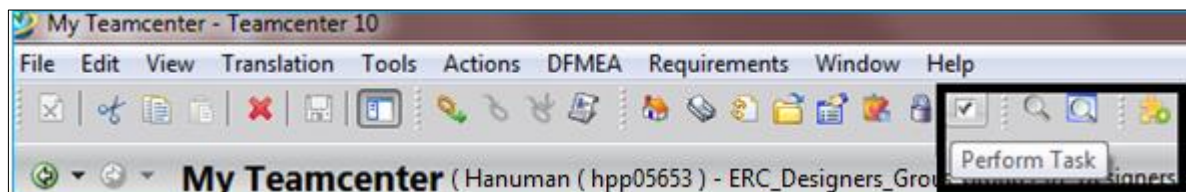


Assign DMU Reviewer (Tool → Assign Participants) by selecting DML object not DML task
Group: DML_Participants_Group, Role: DMU_Reviewer

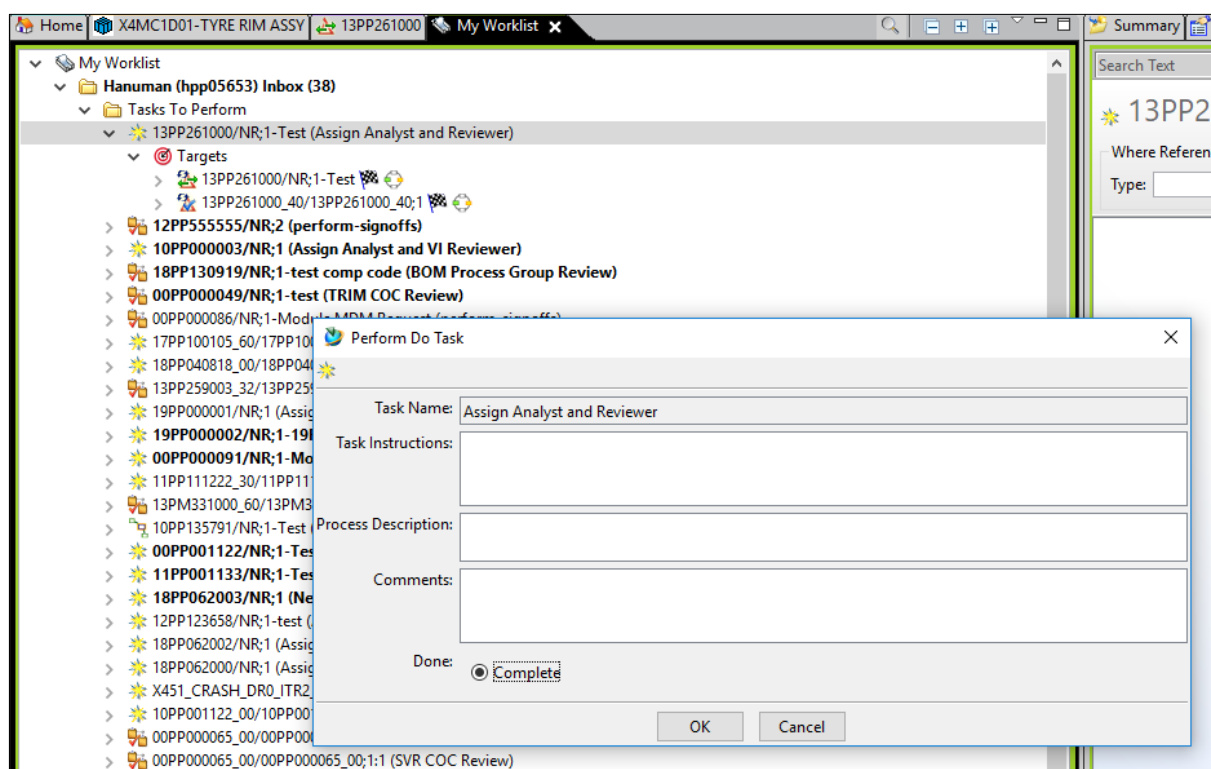


2.4: Perform Task form Analyst worklist (Sign-off)

2.3.1: Do Perform task by selecting 'Assignment' and click perform task



2.3.2: Enter comments if required and click ok.



DML will go to **VI DMU_Reviewer** login to ensure VF update requirement with CoC's

My Worklist

- Karamchand Kalet (kkk04778) Inbox (13)
 - Tasks To Perform
 - Change Request Life Cycle : 25CU123000/NR;1-Test UVF DML TZ 1.85 sanjoy (UVF VI DMU review)
 - Targets
 - 25CU123000/NR;1-Test UVF DML TZ 1.85 sanjoy
 - 25CU123000_30/A;1-Test UVF DML TZ 1.85 sanjoy
 - 22PUA01208_16/A;1-Hybrid Vehicle Release_01_08:4 (Add solution items)
 - 22JU001025_21/A;1-t2 (Add solution items)
 - ERC Released New : 2829H2C0226001/NR;1/2829H2C0226001 (New Add Status Task 1)
 - New Part Request Lifecycle : 5442D1A0116_140625_120925/NR;1 (MCC Review)
 - Module Release Workflow : 13PM327000 (New Released 1)
 - 22JU001003_16/A;1:6 (Add solution items)
 - 22JU001003_00/A;1:5 (Add solution items)
 - 5062D1A0562_081024_163810/NR;1 (MCC Review)
 - 5062D1A0562_081024_163810/NR;1 (MCC Review)
 - 5062D1A0562_081024_163810/NR;1 (MCC Review)
 - 00PP000713/NR;1-GATE MATURATION DML (Assign Analyst and Change Review Board)
 - 22PP081122/NR;1-Test DML (Assign Analyst and Reviewer)
 - Tasks To Track
 - Schedule Tasks
 - Subscription Messages

UVF VI DMU review

Name: UVF VI DMU review Type: Perform Signoff Task

Overview Audit Logs

Properties

Task Instructions: Signoff team perform their signoffs.

Responsible Party: Sourav Karak

Due Date: [Dropdown]

Start Date: 04-Sep-2025

End Date: [Dropdown]

Comments:

Task Result: Unset

Task State: Started

Initiator: Sourav Karak (souravk.ttl)

Group ID: ERC Designers Group

Last Modifying User: Sourav Karak (souravk.ttl)

Actions

Perform Assign...

Perform Signoff

Process: Change Request Life Cycle : 25CU123000/NR;1-Test UVF DML TZ 1.85 sanjoy State: Started

Process Description:

Review Task: UVF VI DMU review

Quorum: 1 Require full participation: false

Responsible Party: Sourav Karak (souravk.ttl)

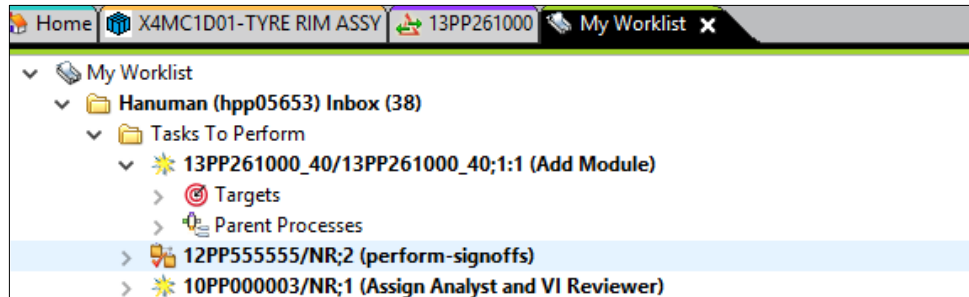
For Review: Attachments All Comments

Instructions: Signoff team perform their signoffs.

User-Group/Role	Required	Decision	Comments	Date
Abhijeet Pawar (amp664685)-DML Particip...	No	No Decision		
Ariun Paithane (amp520167)-DML Particip...	No	No Decision		
Abhijeet Pawar (amp664685)-DML Particip...	No	No Decision		
Karamchand Kalet (kkk04778)-DML Particip...	No	No Decision		
Ashvin Dudhat (ard661971)-DML Participa...	No	No Decision		

Close

Once VI DMU Reviewer approves DML, then DML task will come into Analyst login for assignment 'Add Module'

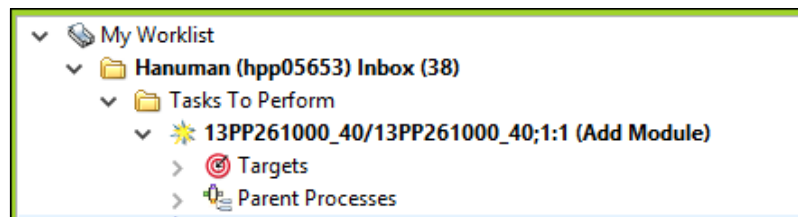


Step 3: Add module

In below conditions variant formula DML shall be used:

- To update variant formula of existing released modules of the same platform: Follow below process
- To update variant formula of carry over released modules in other platform: For this Designer has to copy required modules to new platforms architecture node and follow below process.

Now DML available with CoC's login for assignment 'Add module'. This assignment for attach released modules which you want to update variant formula. Do not attach WIP modules/parts to the Module (VF Change) folder

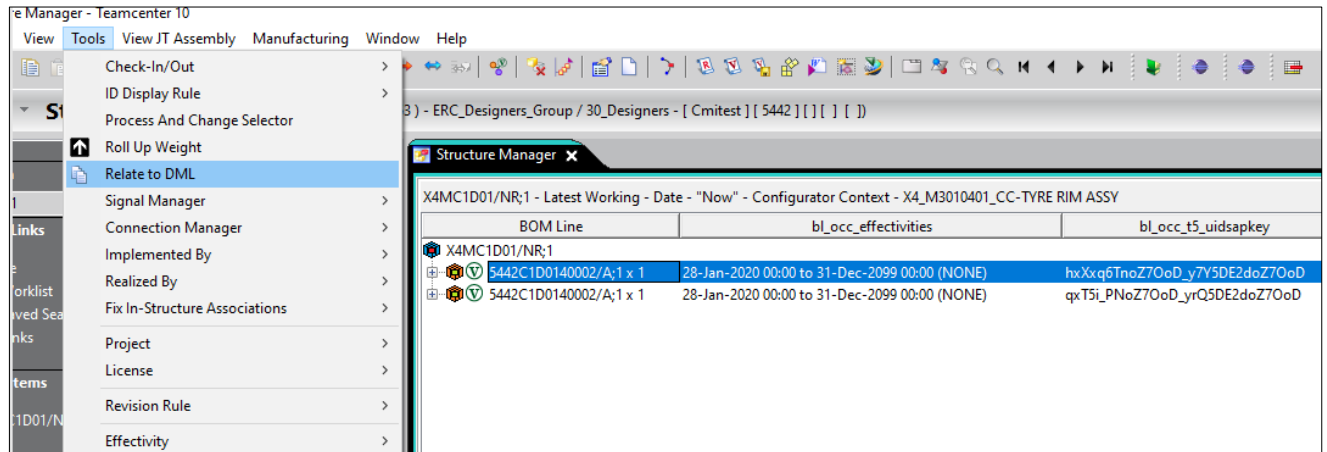


Step 3.1: Go to My Worklist of Analyst - Check Assignment

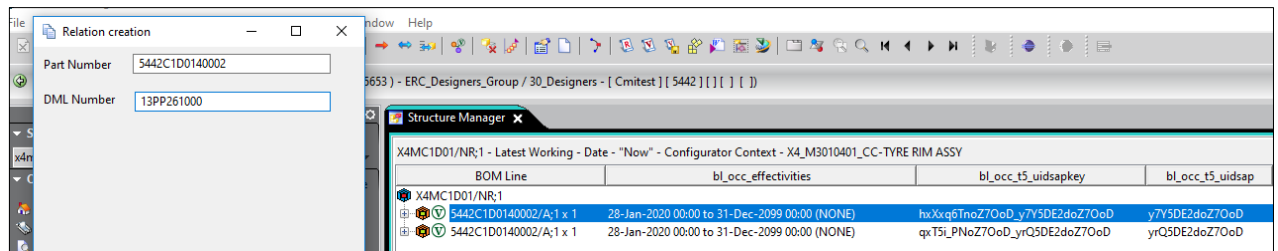
Step 3.2: To attach module to Module (VF Change) folder, open respective architecture node into structure manager.

X4MC1D01/NR;1 - Latest Working - Date - "Now" - Configurator Context - X4_M3010401_CC-TYRE RIM ASSY		
BOM Line	bl_occ_effectivities	bl_occ_t5_uidsapkey
X4MC1D01/NR;1		
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	hxXxq6TnoZ7OoD_y7Y5DE2doZ7OoD
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	qxT5i_PNoZ7OoD_yrQ5DE2doZ7OoD

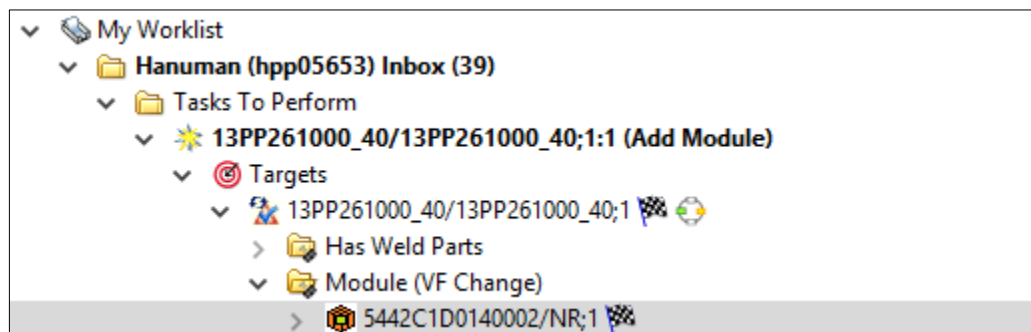
Step 3.3: Select Module occurrence,
Go to Tools -> Relate to DML



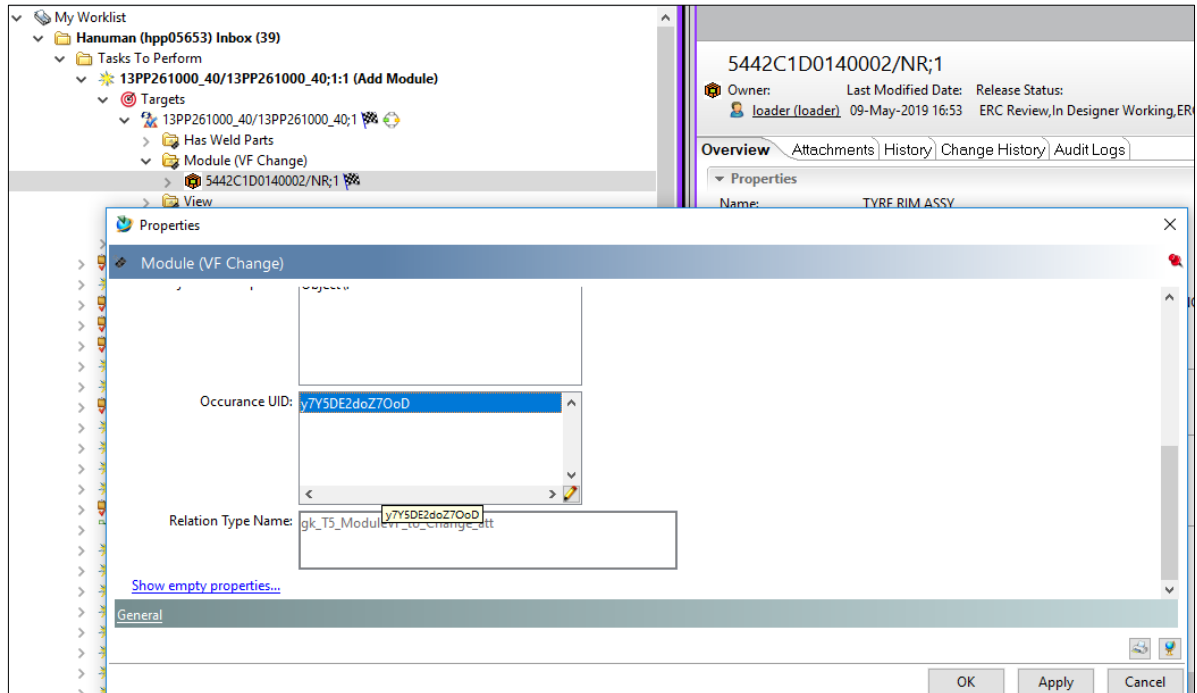
Step 3.4: Enter DML no and click on 'Relate to DML' button and close structure manager



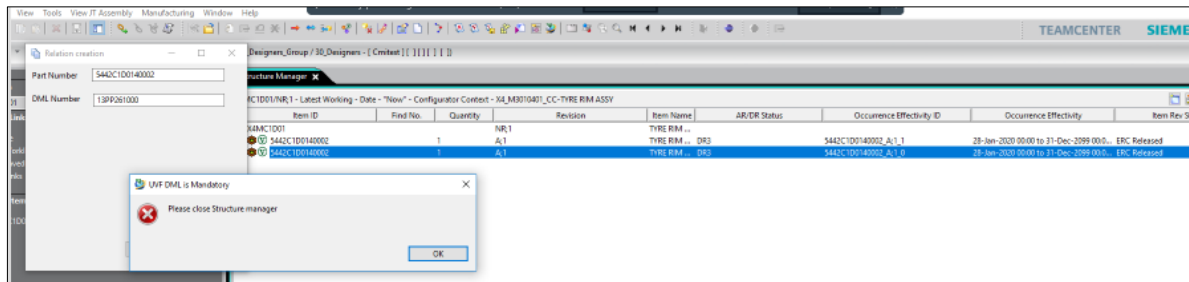
Module occurrence will get Module (VF Change) folder



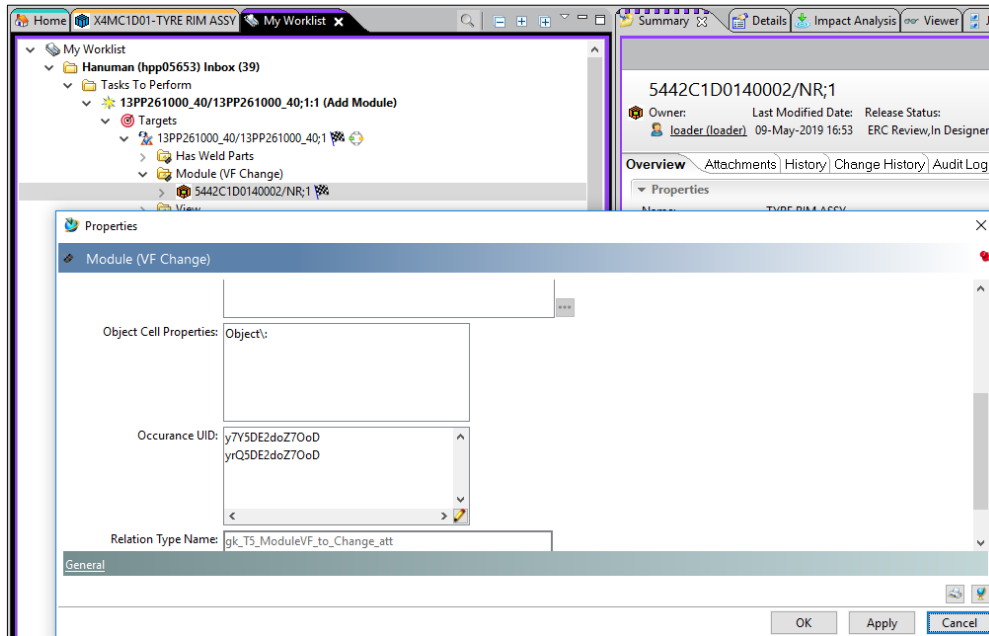
Step 3.5: To view occurrence UID of module, select module -> Right click -> Properties



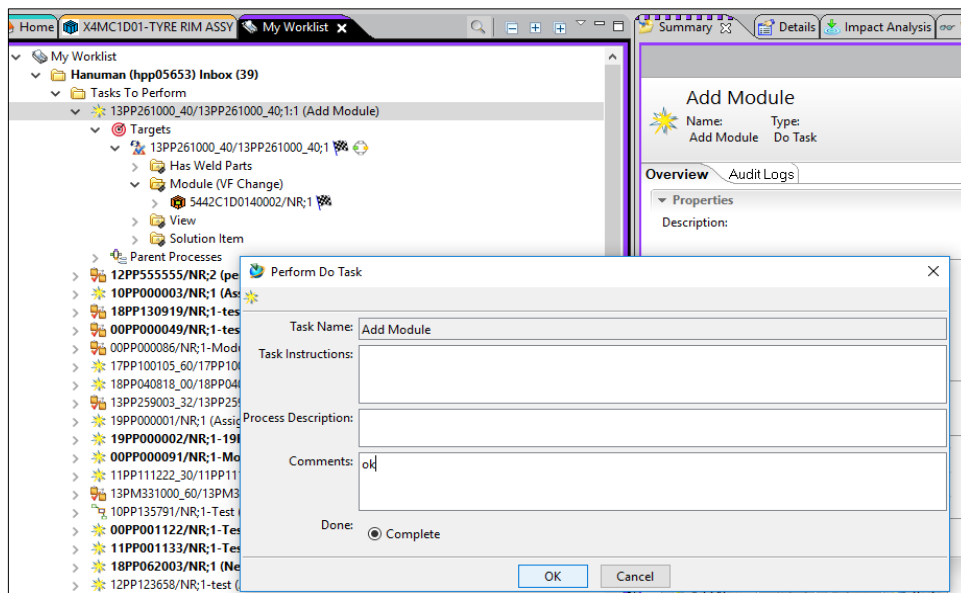
Step 3.6: Attach required occurrences and close structure manager



Step 3.7: Confirm occurrence UID



Step 3.8: Sign-off assignment : After attaching all module occurrences, Select assignment -> Perform task -> OK

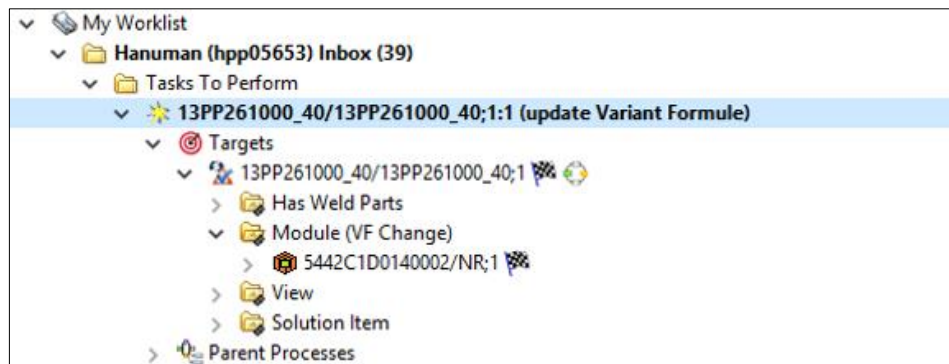


Step 3.9: New occurrence created with blank variant formula. Occurrence effectivity closed of existing occurrences and updated as Today's date to 31-Dec-2099 for new occurrences

Structure Manager - X4MC1D01/NR;1 - Latest Working - Date - "Now" - Configurator Context - X4_M3010401_CC-TYRE RIM ASSY									
BOM Line	bl_occ_effectivities	bl_occ_t5_uisapkey	bl_occ_t5_uisap	Variant Formula	bl_occ_t5_upd_vf_form /	Relative Transformation Matrix	EOC - E...	Release	
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	y7Y5DE2doZ7OoD_0H5DIhoZ7OoD	0H5DIhoZ7OoD	False	100000100-1000001	True	ERC Released		
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	yrQ3DE2doZ7OoD_ECX3DIhoZ7OoD	ECX3DIhoZ7OoD	False	10000100001000001	True	ERC Released		
5442C1D0140002/A;1 x 1	25-Jan-2020 00:00 to 28-Jan-2020 00:00 (NONE)	hxXxq6TnoZ7OoD_y7Y5DE2doZ7OoD	y7Y5DE2doZ7OoD	(((Teamcenter)AGGR...	100000100-1000001	False	ERC Released		
5442C1D0140002/A;1 x 1	25-Jan-2020 00:00 to 28-Jan-2020 00:00 (NONE)	qxT5L_PNoZ7OoD_yrQ3DE2doZ7OoD	yrQ3DE2doZ7OoD	(((Teamcenter)AGGR...	10000100001000001	False	ERC Released		

Step 4: Update variant formula

Now DML come to analyst login for 'update variant formula' assignment



Step 4.1: Send architecture into structure manager and update variant formula of new occurrences

BOM Line	bl_occ_effectivities	bl_occ_t5_uisapkey	bl_occ_t5_uisap	Variant Formula	bl_occ_t5_upd_vf_form /	Relative Transformation Matrix	EOC - E...	Release
X4MC1D01/NR;1							True	
5442C1D0140002/A;1 x 1	25-Jan-2020 00:00 to 28-Jan-2020 00:00 (NONE)	hxXxq6TnoZ						
5442C1D0140002/A;1 x 1	25-Jan-2020 00:00 to 28-Jan-2020 00:00 (NONE)	qxT5L_PNoZ						
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	y7Y5DE2doZ						
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	yrQ3DE2doZ						

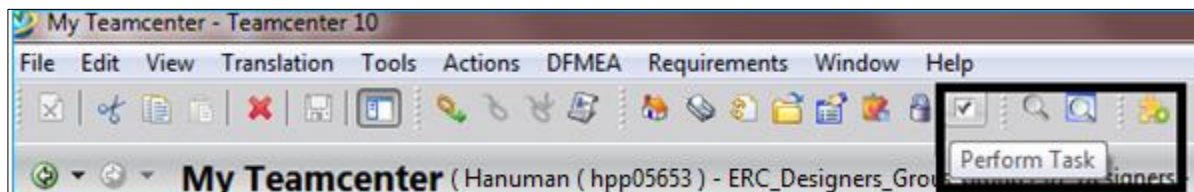
Option	5442C1D014	5442C1D014
[Teamcenter]BALANCE WEIGHT		
PRESS FITTED		
STICK ON TYPE		
[Teamcenter]RIM SIZE		
4.5J x 14		
6.5J x 16		
[Teamcenter]RIM TYPE		
ALLOY		
STEEL		
[Teamcenter]TYRE SIZE, TYPE		
165/80 R14		
195/55 R16		

Step 4.2: SAVE updated variant formula

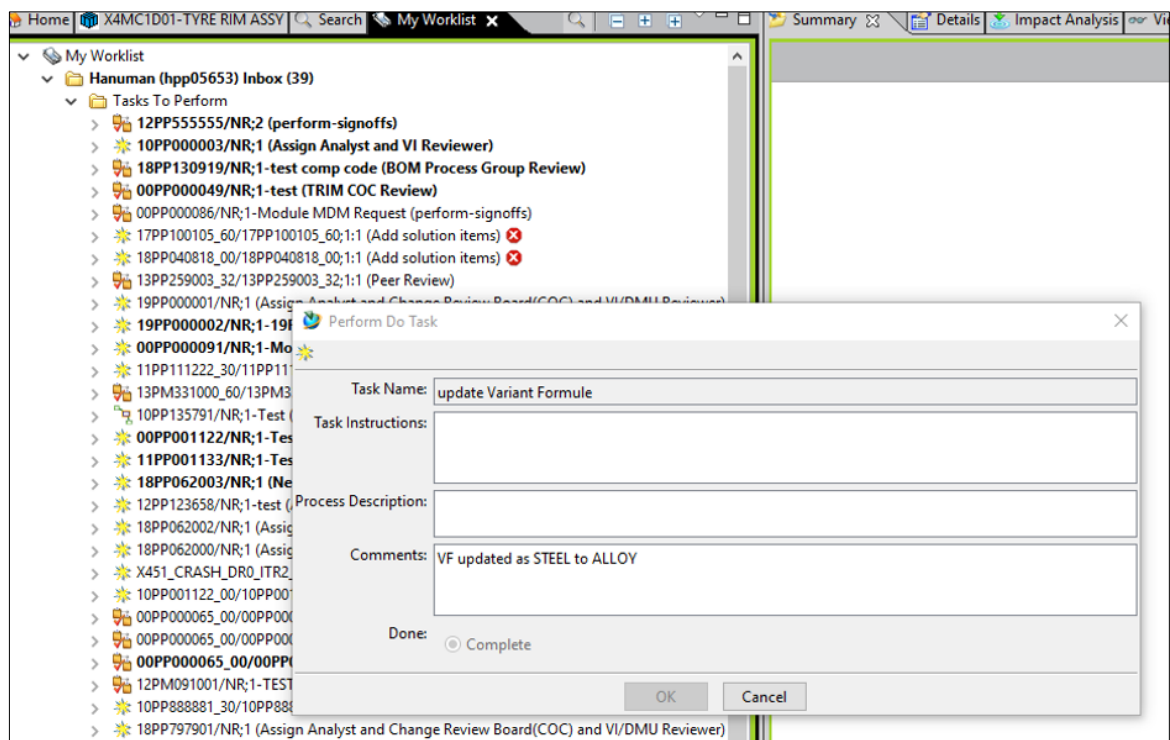
BOM Line	bl_occ_effectivities	bl_occ_15_uidsapkey	bl_occ_15_uidsap	Variant Formula
X4MC1D01/NR;1				
5442C1D0140002/A;1 x 1	25-Jan-2020 00:00 to 28-Jan-2020 00:00 (NONE)	hxXq6TnoZ70oD_y7Y5DE2doZ70oD	y7Y5DE2doZ70oD	(([[Teamcenter]AGGREGATE = CHASSIS AND [Teamcenter]RIM TYPE = STEEL AND [Teamcenter]WHEEL COVER TYP...
5442C1D0140002/A;1 x 1	25-Jan-2020 00:00 to 28-Jan-2020 00:00 (NONE)	qxT5i_PNoZ70oD_ynQ5DE2doZ70oD	ynQ5DE2doZ70oD	(([[Teamcenter]AGGREGATE = CHASSIS AND [Teamcenter]RIM TYPE = STEEL AND [Teamcenter]WHEEL COVER TYP...
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	y7Y5DE2doZ70oD_0t5DIhoZ70oD	0t5DIhoZ70oD	(([[Teamcenter]AGGREGATE = CHASSIS AND [Teamcenter]RIM TYPE = ALLOY AND [Teamcenter]WHEEL COVER TY...
5442C1D0140002/A;1 x 1	28-Jan-2020 00:00 to 31-Dec-2099 00:00 (NONE)	yrQ5DE2doZ70oD_ECK5DIhoZ70oD	ECK5DIhoZ70oD	(([[Teamcenter]AGGREGATE = CHASSIS AND [Teamcenter]RIM TYPE = ALLOY AND [Teamcenter]WHEEL COVER TY...

Variant formula is updated.

Step 4.3: Sign-off the DML by selecting assignment -> Perform task



Step 4.4: Update comment if required and click on 'OK' button



TATA MOTORS	Variant formula updation workflow	NO: ERC/ BPS/ UG/ 01
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Step 5: Review the Changes

DML will come into Reviewers login for assignment “Review the changes”

- 5.1 Reviewer (Manager) will check updated variant formula
- 5.2 After checking reviewer will complete assignment by selecting assignment

Step 6: Acknowledge

DML will come into creators login for assignment ‘Acknowledgement’

After acknowledgement DML will get released and variant formula get updated

Issue No.03	Date: 16-09-2025	TATA MOTORS LIMITED
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